1. File Operations

What is a File?

- A file is used to store data permanently.
- Python can read, write, and update files.

File Modes

- `r` → Read file
- `w` → Write file (old data is deleted)
- `a` → Append (adds new data)
- `x` → Create a new file

`read()`

- Reads the **entire file**.

```
file.read()
```

`readline()`

- Reads **one line at a time**.
- Every call reads the next line.

```
file.readline()
```

Reading Line by Line

```
for line in file:
    print(line)
```

`strip()`

Purpose

- Removes extra spaces.
- Removes newline (`\n`).

```
line.strip()
```

Why use it?

- To print clean output without blank lines.

`write()`

- Writes data into a file.
- Overwrites existing data.

```
file.write("Hello")
```

Append Mode (`a`)

- Adds new data.
- Does not delete existing data.

```
file = open("data.txt","a")
```

`close()`

- Closes the file.
- Always close the file after use.

```
file.close()
```

2. Pandas

What is Pandas?

- Pandas is a Python library.
- Used for data analysis.
- Works with rows and columns.

```
import pandas as pd
```

DataFrame

What is a DataFrame?

- A 2-dimensional table.
- Contains rows and columns.
- Similar to an Excel sheet.

Example

ID	Name	Age
1	John	22

3. Modules, Library, Package
Module

- A single Python file (`.py`).
- Contains functions, variables, or classes.

Example
```
import math
```

Library

- Collection of modules.
- Provides many functionalities.

Examples

- Pandas
- NumPy
- Matplotlib

Package

- Folder containing multiple modules.
- Used to organize code.

Difference
Module

- One Python file.

Library

- Collection of modules.

Package

- Folder containing modules.

4. Exception Handling
What is an Exception?

- An error that occurs while the program is running.

Examples

- Divide by zero
- File not found
- Invalid input

`try`

- Write code that may cause an error.

`except`

- Handles the error.
- Prevents program from crashing.

Why use `try-except`?

- Prevents program termination.
- Handles errors gracefully.
- Improves program reliability.

`finally`

- Always executes.
- Runs whether an error occurs or not.

Used for

- Closing files
- Closing database connections
- Cleaning up resources

5. SQLite Database
What is SQLite?

- Lightweight database.
- Stores data in a `.db` file.
- No separate server required.

Import

```
import sqlite3
```

Connect

```
sqlite3.connect("student.db")
```

- Creates or connects to the database.

Cursor

```
cursor = conn.cursor()
```

Purpose

- Executes SQL queries.

`execute()`

- Executes SQL statements.

Examples

- `CREATE`
- `INSERT`
- `UPDATE`
- `DELETE`
- `SELECT`

```
cursor.execute(sql_query)
```

`commit()`

- Saves changes permanently.

```
conn.commit()
```

Use after

- INSERT
- UPDATE
- DELETE

`fetchall()`

- Retrieves all records.

```
cursor.fetchall()
```

`close()`

- Closes the database connection.

```
conn.close()
```